

Continuous Flow Total Synthesis of Rufinamide

Ping Zhang,[†] M. Grace Russell,[‡] and Timothy F. Jamison^{*,†}[†]Department of Chemistry, Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, Massachusetts 02139, United States[‡]Franciscan University of Steubenville, 1235 University Boulevard, Box 1012, Steubenville, Ohio 43952, United States Supporting Information

ABSTRACT: Small molecules bearing 1,2,3-triazole functionalities are important intermediates and pharmaceuticals. Common methods to access the triazole moiety generally require the generation and isolation of organic azide intermediates. Continuous flow synthesis provides the opportunity to synthesize and consume the energetic organoazides, without accumulation thereof. In this report, we described a continuous synthesis of the antiepileptic medication rufinamide. This route is convergent and features copper tubing reactor-catalyzed cycloaddition reaction. Each of the three chemical steps enjoys significant benefits and has several advantages by being conducted in flow. The total average residence time of the synthesis is approximately 11 min, and rufinamide is obtained in 92% overall yield.

■ INTRODUCTION

Heterocycles containing 1,2,3-triazole ring systems are well-known to possess a variety of biological activities, including anti-HIV,¹ antiallergic,² antifungal,³ antiviral,⁴ and antimicrobial.⁵ An important example is the antiepileptic agent rufinamide (brand name Banzel⁶ or Inovelon,⁷ developed by Novartis and manufactured by Eisai), which was recently approved to treat Lennox-Gastaut syndrome. Pioneered by Huisgen⁸ and further developed by both Sharpless et al.⁹ and Meldal et al.,¹⁰ the 1,3-dipolar cycloaddition reaction between azides and alkynes is the most commonly used method for the construction of 1,2,3-triazole cores. Although often easily accessible, organoazides are generally energetic substances, prone to detonation under even slight energy input from external sources (e.g., heat, light, pressure, etc.). During azide synthesis, toxic and explosive hydrazoic acid may also be generated. For these reasons, the synthesis, handling, isolation, and accumulation of organic azides in larger scale contexts engenders important safety considerations or limits their utility.

The multistep synthesis of complex small molecules in one continuous flow has been an emerging field over the past few years.¹¹ Post-synthetic continuous purification and crystallization of pharmaceutical ingredients and intermediates have also been demonstrated.¹² We envisioned that the safety hazards associated with the production of 1,2,3-triazole compounds can be reduced through the use of continuous flow technology. Herein, we report the first example of continuous total synthesis of rufinamide from readily available starting materials,

featuring a copper-tubing-catalyzed [3 + 2] Huisgen cycloaddition.

■ BACKGROUND

Due to its importance as an antiepileptic medication, several routes have been reported for the synthesis of rufinamide.^{13–17} As shown in Scheme 1, all the routes share a common strategy of the 1,3-dipolar Huisgen cycloaddition of 2,6-difluorobenzylazide (**1**) with a dipolarophile to construct the triazole core. The dipolarophile can be 2-chloroacrylonitrile (**2**, route a),¹³ propiolic acid and esters (route b),¹⁴ or (*E*)-methyl 3-methoxyacrylate (**3**, route c).¹⁵ Upon cycloaddition, subsequent hydrolysis (route a) or amidation (routes b and c) affords rufinamide (**6**). Since the cycloaddition of azide **1** and ester **3** features low-cost starting material and free of catalyst (vs Cu in route b), it has also been demonstrated in continuous flow by Hessel and Noël recently.¹⁶ Regardless of the dipolarophile used, in most of the synthetic routes and production practices reported to date, the energetic intermediate benzyl azide **1** was isolated.¹⁷

We envisioned that, in one continuous flow, rufinamide could be synthesized from the [3 + 2] cycloaddition of **1** and propiolamide (**7**, Scheme 2) without isolation of either intermediate. Both cycloaddition precursors can be generated continuously as well, from commercially available benzyl bromide **8** and methyl propiolate (**9**), respectively. This route would thus feature: (1) Continuous synthesis and immediate consumption of energetic benzyl azide (**1**); (2) *in situ* generation of propiolamide,¹⁸ an expensive reagent prone to polymerization; (3) copper-tubing catalysis of the [3 + 2] cycloaddition; (4) high productivity of rufinamide in a small footprint.

■ RESULTS AND DISCUSSION

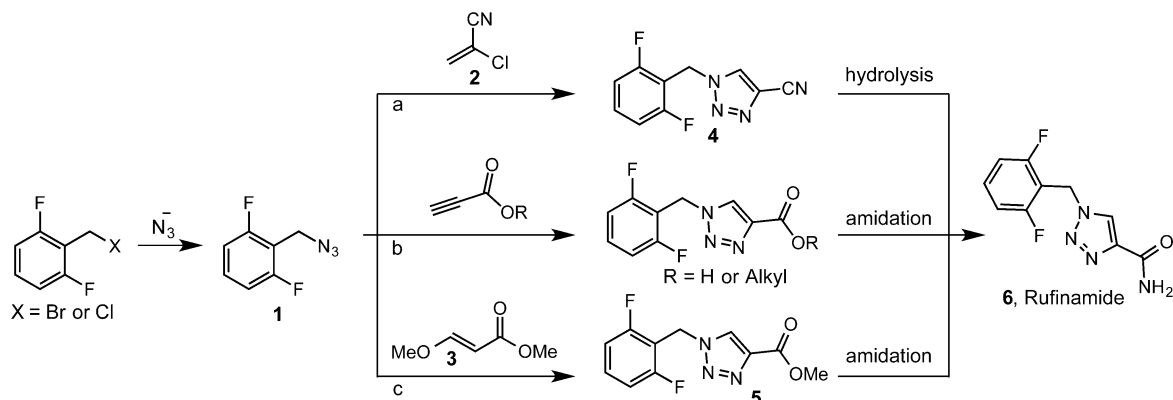
We first investigated the synthesis of 2,6-difluorobenzylazide (**1**) in continuous flow. S_N2 substitution of benzyl halide with azide nucleophiles is known to be a robust transformation to afford **1**.¹⁹ In order to perform the reaction in flow, we evaluated solvents and concentration, followed by temperature and reaction residence time. As shown in Scheme 3, under the optimal condition, a 1 min average residence time was sufficient for the synthesis of benzyl azide **1** at room temperature from

Special Issue: Continuous Processes 14

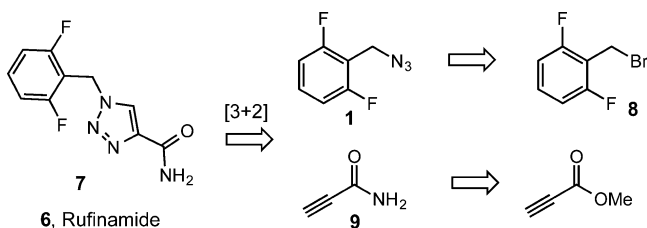
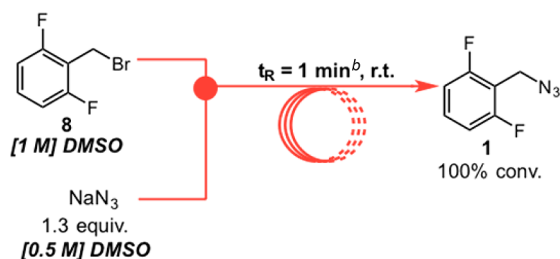
Received: May 27, 2014

Published: August 27, 2014

Scheme 1. Overview of the synthetic routes towards rufinamide (6)



Scheme 2. Retrosynthetic analysis for the continuous-flow synthesis of rufinamide (6)

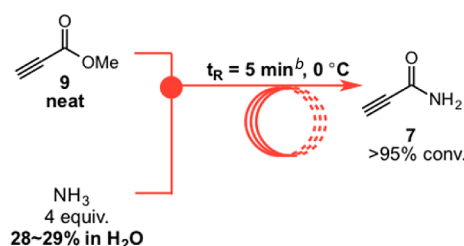
Scheme 3. Continuous-flow synthesis of 2,6-difluorobenzylazide^a

^aThe overall reaction concentration was [0.29 M]. ^bThe residence time was estimated based on pumping flow rates of the two feeds.

benzyl bromide **8**.²⁰ Presumably, the short reaction time is resulted from efficient mixing in flow. We identified DMSO as the best solvent, both for solubility and for reactivity.

Next, we examined the possibility of synthesizing propiolamide (**7**) in flow.²¹ Both methyl propiolate and propiolamide are prone to polymerization;^{18,22} therefore, the reaction condition were evaluated to enhance the desired amidation while minimizing polymerization. As shown in Scheme 4, the optimum conditions that we developed to provide propiolamide (**7**) in flow are as follows: 4 equiv of ammonium hydroxide, 0 °C (vs −78 °C in batch), and 5 min average residence time. Under these conditions, 95% conversion of methyl propiolate (**9**) to propiolamide was obtained. A higher temperature or longer reaction time resulted in increased byproduct formation. Worth noting also is the high volumetric throughput of this reaction: the only solvent was water from the ammonium aqueous solution.

As the first two reactions were developed in flow, we explored the linkage of the two for the subsequent [3 + 2] cycloaddition (Scheme 5). Mixing the two reaction-streams did not cause any solid formation; the reaction mixture remained

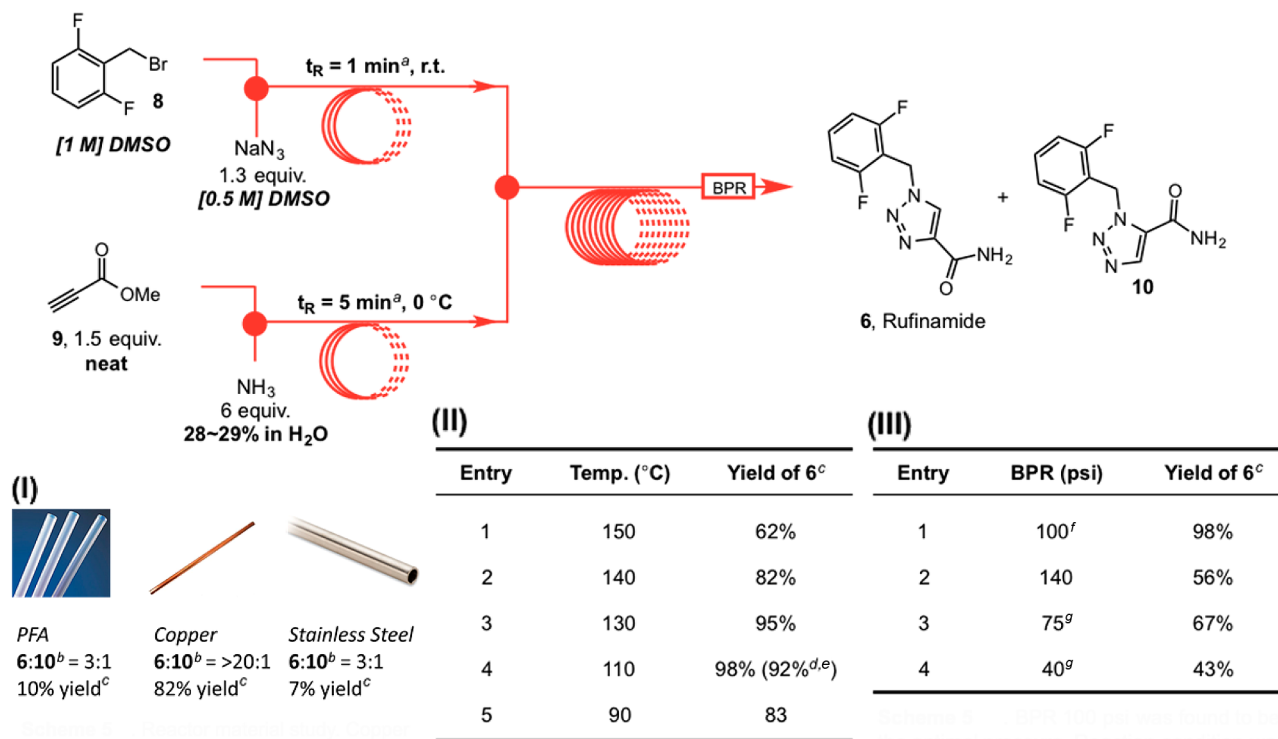
Scheme 4. Continuous-flow synthesis of propiolamide (**7**) from methyl propiolate (**9**)^a

^aThe overall reaction concentration was [2.8 M]. ^bResidence time was estimated based on pumping flow rates of the feeds.

homogeneous. The rate of the desired cycloaddition, however, was very poor. With a residence time of 6.2 min at 140 °C, only 10% yield of rufinamide was detected by HPLC (Scheme S(1), PFA tubing employed). Formation of the 1,5-regioisomer **10** was also observed. Copper salts are known to catalyze this transformation, to improve both reactivity and regioselectivity.²³ More recently, commercial copper tubing has been used to catalyze a variety of reactions including click chemistry, likely occurring by the release of trace amounts of ionic copper species into the reaction streams.²⁴ Specifically, Bogdan and Sach demonstrated a triazole synthesis in copper tubing with in situ generation of organic azide species.^{24b} Therefore, we examined copper tubing as a reactor material in this context in the rufinamide synthesis (Scheme S1, copper tubing). Under otherwise identical reaction conditions, we observed an 82% overall yield of rufinamide, with greater than 20:1 regioselectivity. Similar enhancement of this reaction was not observed when stainless steel tubing was instead used (Scheme S1, stainless steel tubing).

In order to improve the yield of rufinamide further, the reaction temperature and pressure were investigated. As shown in Scheme S(II), we identified the optimal temperature to be 110 °C (entry 4). Under these conditions, the overall yield of rufinamide was 98% by HPLC and the isolated yield 92%. We hypothesize that, at higher temperatures (entries 1–3), decomposition of the benzyl azide (**1**) begins to compete with the desired sequence, thus resulting in lower yield. Similar observations have also been reported by Hessel and Noël.¹⁶ On the other hand, when the temperature was under 90 °C (entry 5), the rufinamide yield was lower, likely due to less copper release into the reaction stream or a slower reaction rate at the lower temperature.

Scheme 5. Reaction condition study for the continuous-flow synthesis of rufinamide (6)



(I) Reactor material study. Copper tubing manufactured material proved to be optimal. Reaction condition: 140 °C reaction temperature, 100 psi BPR, 6.2 min residence time. (II) 110 °C was found to be the optimal reaction temperature. Reaction conditions were: copper tubing reactor, 100 psi BPR, 6.2 min residence time. The overall reaction concentration was estimated to be [0.25 M]. (III) BPR 100 psi was found to be the optimal pressure. Reaction conditions were: copper tubing reactor, 100 psi BPR, 6.2 min residence time. The overall reaction concentration was estimated to be [0.25 M]. ^aResidence time was estimated based on pumping rate of feeds. ^bNMR ratio. ^cHPLC yield of the three steps. Biphenyl was used as an internal standard. ^dIsolated yield. ^eProductivity based on this yield was 217 mg/h. ^fMinor gassing observed. ^gSignificant gassing observed.

During the course of evaluating the reaction temperature (100 psi BPR), we also observed occasional gas generation. While small amounts of gas release may reduce the residence time, we investigated how increased pressure would affect the process, with an eye toward suppressing this phenomenon and maintaining a steady residence time. One surprising finding was that, although no off-gassing was observed with a 140 psi BPR, the reaction outcome was quite poor: Only 56% yield of rufinamide was observed (Scheme SIII, entry 2). We thus hypothesized that the gas in reactor was ammonia. When ammonia dissolves well under high pressure (e.g., 140 psi), it complexes copper and interferes with catalysis. While under lower pressure (e.g., 100 psi), some of ammonia escapes from solution, leaving copper remaining active. Further decreasing the back pressure to 75 psi (entry 3) and 40 psi (entry 4) resulted in poor yield as well, due to reduced residence time resulting from significant gas generation.

CONCLUSIONS

A continuous total synthesis of rufinamide in flow has been developed. Many advantages have been demonstrated to overcome the limitations that would have been imposed by conducting an analogous synthesis in batch. The safety hazards were significantly minimized because an organoazide intermediate was neither accumulated nor isolated. The otherwise costly and unstable propiolamide was prepared in-line and could be employed without concerns of polymerization or storage and also without the need for further functional group manipulation because the final target was the direct product.

The overall average residence time was 11 min, and rufinamide was afforded in 92% overall yield and high selectivity.

EXPERIMENTAL SECTION

General Information. Parts for the construction of flow reactors are purchased from IDEX Health & Science Technologies, unless otherwise stated. Regular flow reactors were constructed from high purity PFA tubing with 1/16" o.d. and 0.03" i.d. and PEEK 1/4–28 nut. Copper flow reactors were constructed from copper C122 seamless tubing with 1/16" o.d. and 0.03" i.d., purchased from Amazon.com (supplier: Small Parts) and stainless steel 1/16" female nut purchased from Swagelok. Stainless steel flow reactors were constructed from stainless steel 316 tubing with 1/16" o.d. and 0.03" i.d. and stainless steel 1/16" female nut purchased from Swagelok. The metal reactors were linked with PFA tubing through 1/16" stainless steel unions purchased from Swagelok. The reagents and reaction streams were mixed with Tefzel T-mixers with 0.02" thru. The reaction pressure was controlled by back-pressure regulators (BPRs). Reagents were delivered by either PhD Ultra syringe pumps purchased from Harvard Apparatus equipped with high-pressure stainless steel syringes and SGE airtight glass syringes purchased from VWR or Asia syringe pumps from Syrris.

Chemicals and solvents were purchased from Sigma-Aldrich and used without further purification unless otherwise stated. 2,6-Difluorobenzyl bromide was purchased from Oakwood Chemical.

Continuous Total Synthesis of Rufinamide (6). A DMSO solution of 2,6-difluorobenzyl bromide [1 M] and biphenyl (internal standard) [0.1 M] was loaded in an 8 mL stainless steel syringe (feed A). A [0.5 M] DMSO solution of sodium azide was loaded in a second 8 mL stainless steel syringe (feed B). Neat methyl propiolate (feed C) and ammonium hydroxide (~28% ammonia, feed D) were loaded in 2 mL SGE glass syringes, respectively. All four syringes were pumped with Harvard Apparatus syringe pumps. Feed C was pumped at 2.2 $\mu\text{L}/\text{min}$, and feed D was pumped at 6.6 $\mu\text{L}/\text{min}$. The two feeds were mixed and passed through a 40 μL PFA reactor; both mixer and reactor were cooled in a ice–water bath. At the meanwhile, feed A was pumped at 16.5 $\mu\text{L}/\text{min}$, and feed B was pumped at 41.3 $\mu\text{L}/\text{min}$ and stream upon mixing was passed through a 57 μL PFA reactor at room temperature. The two outlets were jointed with a T-mixer and passed through a 431 μL copper tubing reactor. The reactor was heated at 110 $^{\circ}\text{C}$ and equipped with a 100 psi back-pressure regulator (BPR). The overall calculated residence time for the continuous-flow reaction sequence was 11 min. The reaction was run for 4-residence-time (44 min) to reach steady state, and the reaction was collected for 60 min and afforded brown/red solution. Two drops of the solution was diluted with methanol to 1 mL, which was then analyzed by LCMS to give 98% yield. To the remaining reaction mixture was added water (two times the volume of reaction mixture) while stirring and let the resulting slurry set for 15 min. Upon filtration, washing with water, the off-white sticky cake was dried in vacuum oven for 24 h. The dried off-white solid afforded 215 mg rufinamide (92% yield). NMR in DMSO- d_6 was in accordance with literature.²⁵ The productivity of this sequence was 217 mg/h.

■ ASSOCIATED CONTENT

■ Supporting Information

Experimental procedures and NMR spectra. This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: tfj@mit.edu; tel.: +1 617-253-2135.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the Defense Advanced Research Projects Agency (DARPA N66001-11-C-4147). The authors would like to thank Prof. Klavs F. Jensen, Prof. Allan S. Myerson, and their research groups and Dr. Chunhui Dai for their helpful discussions and suggestions.

■ REFERENCES

- (1) (a) Alvarez, R.; Valaquez, S.; San, F.; Aquaro, S.; De, C.; Penro, C. F.; Karlsson, A.; Balzarini, J.; Camarasa, M. J. *J. Med. Chem.* **1994**, 37, 4185. (b) Valquez, S.; Alvarez, R.; Perez, C.; Cago, F.; De, C.; Balzarini, J.; Camarasa, M. J. *Antiviral Chem. Chemother.* **1998**, 9, 481.
- (2) Buckle, D. R.; Rockell, C. J. M.; Smith, H.; Spicer, B. A. *J. Med. Chem.* **1986**, 29, 2262.
- (3) (a) Vicentini, C. B.; Brandolini, V.; Guarneri, M.; Giori, P. *Farmaco* **1992**, 47, 1021. (b) Fung-Tomc, J. C.; Huczko, E.; Minassion, B.; Bonner, D. P. *Antimicrob. Agents Chemother.* **1998**, 42, 313.
- (4) Palhagen, S.; Canger, R.; Henriksen, O.; van Paryes, J. A.; Riviere, M. E.; Karolchik, M. A. *Epilepsy Res.* **2001**, 43, 115.
- (5) Genin, M. J.; Allwine, D. A.; Anderson, D. J.; Barbachyn, M. R.; Emmert, D. E.; Garmon, S. A.; Graber, D. R.; Grege, K. C.; Hester, J. B.; Hutchinson, D. K.; Morris, J.; Reischer, R. J.; Ford, C. W.; Zurenko, G. E.; Hamel, J. C.; Schaadt, R. D.; Stapert, D.; Yagi, B. H. *J. Med. Chem.* **2000**, 43, 953.
- (6) In the United States.
- (7) In the European Union.
- (8) Huisgen, R. *Proc. Chem. Soc.* **1961**, 357.
- (9) Rostovtsev, V. V.; Green, L. G.; Fokin, V. V.; Sharpless, K. B. *Angew. Chem., Int. Ed.* **2002**, 41, 2596.
- (10) Tornøe, C. W.; Christensen, C.; Meldal, M. *J. Org. Chem.* **2002**, 67, 3057.
- (11) For selected examples: (a) Bogdan, A. R.; Poe, S. L.; Kubis, D. C.; Broadwater, C. J.; McQuade, D. T. *Angew. Chem., Int. Ed.* **2009**, 48, 8547. (b) Lévesque, F.; Seeberger, P. H. *Angew. Chem., Int. Ed.* **2012**, 51, 1706. (c) Hopkin, M. D.; Baxendale, I. R.; Ley, S. V. *Org. Biomol. Chem.* **2013**, 11, 1822. (d) Newton, S.; Carter, C. F.; Pearson, C. M.; Alves, L.; de, C.; Lange, H.; Thansandote, P.; Ley, S. V. *Angew. Chem., Int. Ed.* **2014**, 53, 4915.
- (12) (a) Snead, D. R.; Jamison, T. F. *Chem. Sci.* **2013**, 4, 2008. (b) Mascia, S.; Heider, P. L.; Zhang, H.; Lakerveld, R.; Benyahia, B.; Barton, P. I.; Braatz, R. D.; Cooney, C. L.; Evans, J. M. B.; Jamison, T. F.; Jensen, K. F.; Myerson, A. S.; Trout, B. L. *Angew. Chem., Int. Ed.* **2014**, 52, 12359. (c) Heider, P. L.; Born, S. C.; Basak, S.; Benyahia, B.; Lakerveld, R.; Zhang, H.; Hogan, R.; Buchbinder, L.; Wolfe, A.; Mascia, S.; Evans, J. M. B.; Jamison, T. F.; Jensen, K. F. *Org. Proc. Res. Dev.* **2014**, 18, 402.
- (13) Portmann, R. Novartis, AG, Basel, Switzerland. U.S. Patent 6,156,907, 2000.
- (14) Acid examples: (a) Attolino, E.; Colombo, L.; Mormino, I.; Allegrini, P. Dipharma Francis S.R.L., Baranzate, Italy. US Patent 2010/0234616 A1, 2010. (b) Attolino, E.; Colombo, L.; Mormino, I.; Allegrini, P. Dipharma Francis S.R.L., Baranzate, Italy. US Patent 8,198,459 B2, 2012. (c) De Leon Martin, A. A.; Bellmunt, J. B.; Clotet, J. H.; Carandell, L. S.; Pascual, G. F.; Bertran, J. C.; Barjoan, P. D. Laboratorios Lesvi, S.L., Spain. US Patent 2013/0045998 A1, 2013. Ester examples: (d) Kankan, R. N.; Rao, D. R.; Birari D. R. Cipla Limited, India. WO Patent 2010/043849, 2010. (e) Kankan, R. N.; Rao, D. R.; Birari D. R. Cipla Limited, India. US Patent 8,183,269 B2, 2012.
- (15) Mudd, W. H.; Stevens, E. P. *Tetrahedron Lett.* **2010**, 51, 3229.
- (16) Borukhova, S.; Noël, T.; Metten, B.; de Vos, E.; Hessel, V. *ChemSusChem* **2013**, 6, 2220.
- (17) Exceptions: (a) Reference 14c, no concentration to deliver neat azide, but extraction and removal of aqueous waste was conducted. (b) One-pot, propiolamide as dipolariphole: Rajadhyaksha, M. N.; Nair, R.; Ramesan, P. V.; Johnson, K.; Panandikar, A. M. Indoco Remedies Ltd., India. WO Patent 2012/032540 A1, 2012.
- (18) Selected examples: (a) Allison, J. P.; Michel, R. E. *Chem. Commun. (London)* **1966**, 762. (b) MacNulty, B. J. *Polymer* **1966**, 7, 275.
- (19) For a review, see Scriven, E. F. V.; Turnbull, K. *Chem. Rev.* **1988**, 88, 297.
- (20) For a batch condition, see ref 14c.
- (21) Example of batch conditions: Hay, L. A.; Koenig, T. M.; Ginah, F. O.; Copp, J. D.; Mitchell, D. J. *Org. Chem.* **1998**, 63, S050.
- (22) For a selected example: Kuroda, H.; Tomita, I.; Endo, T. *Polymer* **1997**, 38, 6049.
- (23) For selected examples: (a) Himo, F.; Hilgraf, R.; Rostovtsev, V. V.; Noodleman, L.; Sharpless, K. B.; Fokin, V. V. *J. Am. Chem. Soc.* **2005**, 127, 210. (b) Spiteri, C.; Moses, J. E. *Angew. Chem., Int. Ed.* **2010**, 49, 31.
- (24) (a) Zhang, Y.; Jamison, T. F.; Patel, S.; Mainolfi, N. *Org. Lett.* **2011**, 13, 280. (b) Bogdan, A. R.; Sach, N. W. *Adv. Synth. Catal.* **2009**, 351, 849.
- (25) Cherynyak, S.; Cyjon, R.; Ozer, I. Taro Pharmaceutical Industries Limited, Haifa Bay, Italy. US Patent 2014/0155619 A1, 2014.